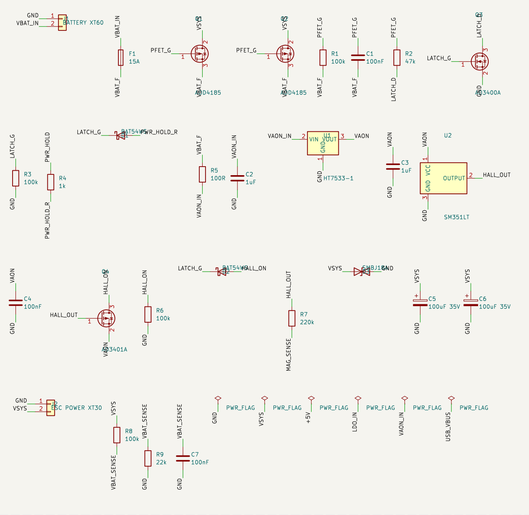
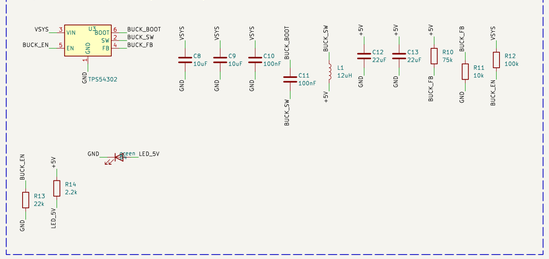
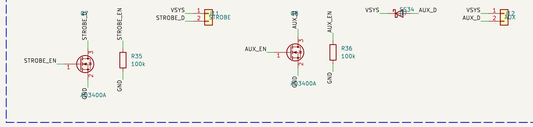
Battery input, 15 A fuse, magnet soft-latch power switch (SM351LT, always-on 3.3 V)



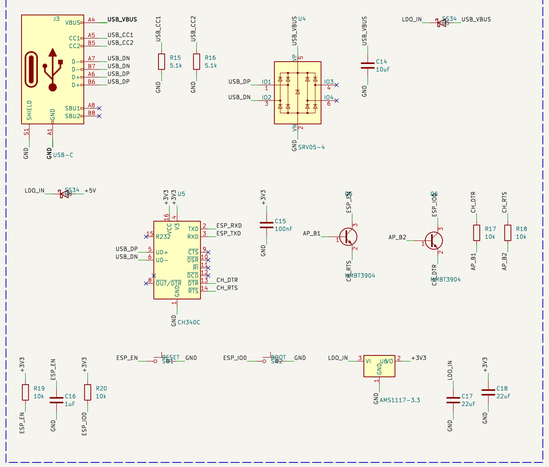
5 V / 3 A buck regulator (TPS54302)



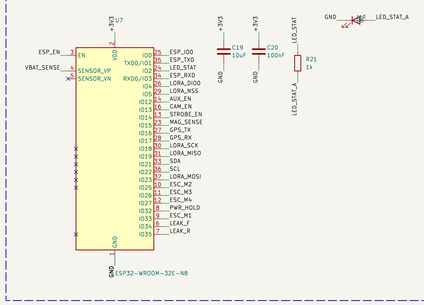
Switched outputs: strobe and spare (lights / pump)



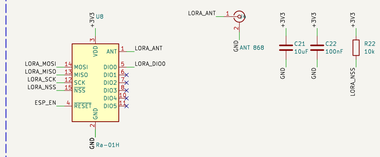
USB-C programming port, CH340C USB serial, auto-program, 3.3 V LDO



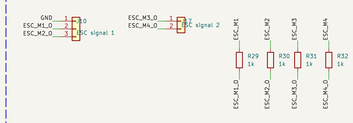
ESP32-WROOM-32E (Wi-Fi to the camera, BLE)



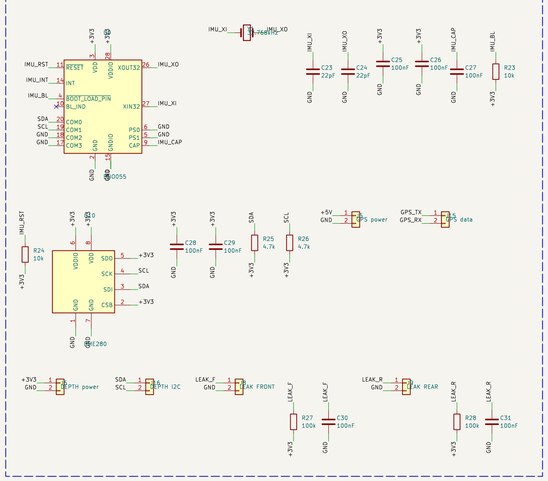
LoRa radio (Ai-Thinker Ra-01H, SX1276, 868 MHz) and antenna socket



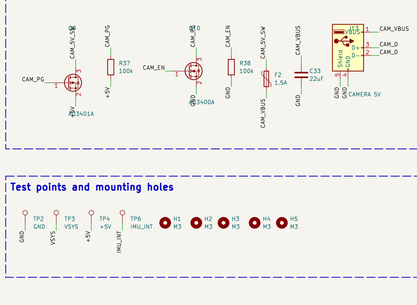
ESC signals (DShot to four single ESCs)



Sensors: BNO055 compass/IMU, BME280 hull monitor, GPS, depth, leak



4K camera: switched 5 V USB-A port



Test points and mounting holes

Generated from design.py by generate_board.py — edit design.py, not this file.
Magnet power switch: a magnet near U2 (SM351LT, always powered by U1, 3 uA) pulls HALL_OUT low -> Q4 drives
LATCH_G high -> Q3 pulls the P-FET gates low -> the ESP32 boots and drives PWR_HOLD (GPIO32) high to stay on.
Shutdown = the firmware releases PWR_HOLD once the magnet is taken away. MAG_SENSE (GPIO15) reads the magnet.
USB-C powers only the 3.3 V side (D5): the shore-station board needs nothing but a USB cable.
All analog inputs are on ADC1 (GPIO34/35/36/39), which keeps working while Wi-Fi talks to the camera.

Single sheet. Nets are connected by name (labels).
All-in-one, JLCPCB-assembled: ESP32, USB-C, magnet power latch, 5V/3.3V, IMU, hull sensor, LoRa

BIOHUB / submarine		
Sheet: /		
File: minnow-core.kicad_sch		
Title: Minnow AUV core board		
Size: A1	Date: 2026-11-11	**Rev: 2.0**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1